

Develop Natural Language Solutions in Azure

Language, speech, and translation with Azure AI Services

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What You'll Learn

- Pull meaning out of text: language, sentiment, entities, key phrases, PII
- Turn speech into text and text into speech
- Decide between a purpose-built service and a large model
- Build a spoken conversation that reacts in real time
- Translate written and spoken language

Problem: Human Language Is Not Data

A support mailbox receives 4,000 messages a week.

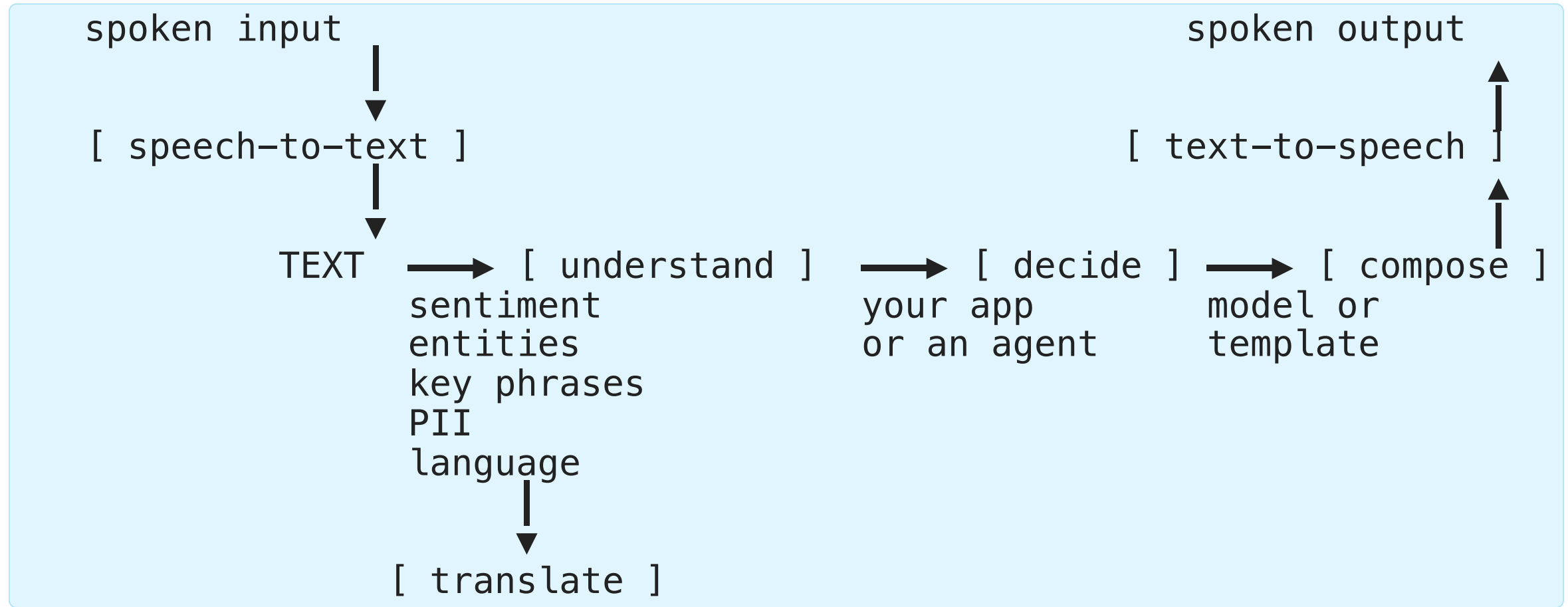
- Which are angry? Which mention a competitor? Which contain a customer's phone number?
- **✗** A database cannot answer any of that with a **WHERE** clause
- **✗** Reading them by hand does not scale

Need: services that convert language into structured facts you can filter, route, and store.

That is the whole point of this presentation: **language in, data out.**

The Pipeline

Every solution in this presentation is a few of these boxes in a row.



Voice Live is this whole loop, running continuously, in under a second.

Decoding the Jargon

Term	In plain words
NER	Named Entity Recognition — find the names of people, places, dates
Entity linking	Also say <i>which</i> Paris — by linking to a knowledge base entry
Key phrases	The handful of noun phrases the text is actually about
PII	Personally Identifiable Information — emails, phone numbers, ID numbers
Redaction	Replacing those values with ***** before you store or send the text
STT / TTS	Speech-to-text (listening) / text-to-speech (talking)
Neural voice	A model-generated voice, e.g. en-US-JennyNeural
SSML	XML that tells the voice where to pause, and in which style to speak
Diarization	Marking <i>who</i> said each sentence when several people speak
VAD	Voice Activity Detection — deciding when the person stopped talking

Purpose-Built Service or Large Model?

Both can read text. They fail in different ways.

Ask yourself	Language / Speech service	Generative model
Do I need the <i>same</i> answer every time?	Yes — fixed categories, scores	No — wording varies
Do I need an open-ended answer?	No	Yes
Do I need a confidence score?	Yes, per result	Not really
Cost and latency	Low and predictable	Higher, varies with length

Rule of thumb: if you can write the list of possible answers in advance, use the purpose-built service.

Azure Language Service

TextAnalyticsClient

Package: `pip install azure-ai-textanalytics`

```
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential
from azure.identity import DefaultAzureCredential

# Key-based
client = TextAnalyticsClient(endpoint="https://{resource}.services.ai.azure.com",
                             credential=AzureKeyCredential("KEY"))

# Recommended (production)
client = TextAnalyticsClient(endpoint="https://{resource}.services.ai.azure.com",
                             credential=DefaultAzureCredential())
```

Constraints: max 5,120 characters per document; max 1,000 items per collection

Language Detection

```
result = client.detect_language(["Hello world", "Bonjour le monde"])
for doc in result:
    lang = doc.primary_language
    print(f"{lang.name} ({lang.iso6391_name}) – confidence: {lang.confidence_score}")
```

Edge cases:

- Multilingual text → returns dominant language with lower confidence
- Unparseable text → `language = "(unknown)", score = 0`

Sentiment Analysis

[EXAM]

```
result = client.analyze_sentiment(["I love this service!", "The wait was too long."])
for doc in result:
    print(f"Overall: {doc.sentiment}")
    print(f"    positive={doc.confidence_scores.positive:.2f}")
    for sent in doc.sentences:
        print(f"    Sentence: '{sent.text}' → {sent.sentiment}")
```

Sentiment logic:

- All neutral → neutral
- Includes positive + neutral → positive
- Includes negative + neutral → negative
- Positive + negative both present → mixed

Entity Recognition and Linking

[EXAM]

Named Entity Recognition (NER):

```
result = client.recognize_entities(["Microsoft was founded by Bill Gates in 1975."])
for doc in result:
    for entity in doc.entities:
        print(f"{entity.text} ({entity.category}) – {entity.confidence_score:.2f}")
```

Categories: Person, Location, DateTime, Organization, Address, Email, URL

Entity Linking:

```
result = client.recognize_linked_entities(["I visited Paris last summer."])
for doc in result:
    for entity in doc.entities:
        print(f"{entity.name} – {entity.data_source} – {entity.url}")
```

Entity linking uses Wikipedia as knowledge base; disambiguates same-name entities by context (e.g., "Venus" → planet vs. mythology)

Custom Named Entity Recognition

[EXAM]

Train a custom entity model when the prebuilt categories do not match your domain.

Problem: a broad `ContactInfo` entity has **low precision** — it matches phone numbers, e-mail addresses, handles, and whatever sits near them.

Fix the definition, not the volume: replace it with specific `Phone`, `Email`, and `SocialMedia` entity types and relabel the training data.

Tempting fix	Why it fails
Lower the confidence threshold	Returns more false positives
Auto-label the training data	Reproduces the same ambiguous definition
Add more training data	More examples of an overly broad entity

Rule: low precision caused by an ambiguous label is a schema problem, not a data-volume problem.

Retrieval Practice — Lab 1

Pause here and complete [Lab 1](#) before continuing.

Personally Identifiable Information (PII)

[EXAM]

Use PII recognition to find and redact sensitive details before storage, indexing, or model input.

```
result = client.recognize_pii_entities([
    "Contact Rick at rick@example.com or 555-0100."
])

for doc in result:
    print(doc.redacted_text)
    for entity in doc.entities:
        print(f"{entity.text} ({entity.category})")
```

The response provides:

- `redacted_text` with detected values replaced
- Entity text, category, offset, length, and confidence score
- Categories such as person, email, phone number, address, and government ID

What masking actually does — the sensitive value is **removed**, not hidden alongside the original, and each value is replaced by its **category label**:

```
Input:  Contact Rick at rick@example.com or 555-0100.
Output: Contact [PERSON] at [EMAIL] or [PHONENUMBER].
```

There is no overarching Contact category. Detection uses specific categories (person, email, phone number), so a policy that must treat phone numbers differently from names can do so.

Pipeline pattern: detect → review policy → redact or tokenize → persist the safe representation.

Do not treat detection as authorization: combine PII analysis with access controls, encryption, retention policies, and audit logging.

Content Safety — Text Analysis

[EXAM]

Check user-generated text **before** it is published.

```
from azure.ai.contentsafety import ContentSafetyClient
from azure.ai.contentsafety.models import AnalyzeTextOptions

request = AnalyzeTextOptions(text=comment)
response = client.analyze_text(request)

for item in response.categories_analysis:
    print(item.category, item.severity)    # e.g. SelfHarm 4
```

Element	Value
Options class	AnalyzeTextOptions, with the input passed as the named text parameter
Client method	client.analyze_text(request) — the synchronous ContentSafetyClient
Result	Per-category severity, including self-harm

Content Safety is not PII detection: it classifies harmful-content severity; it does not find or mask personal data.

Key Phrase Extraction

```
result = client.extract_key_phrases([  
    "Microsoft launched its new AI product in Seattle this week."  
)  
for doc in result:  
    print(doc.key_phrases)    # ['new AI product', 'Microsoft', 'Seattle']
```


Retrieval Practice — Lab 2

Pause here and complete [Lab 2](#), then check its solution.

Language MCP Server

Language MCP Endpoint and Tools

[EXAM]

Endpoint:

<https://{foundry-resource-name}.cognitiveservices.azure.com/language/mcp?api-version=2025-11-15-preview>

Tools exposed as MCP:

Tool	Capability
Named Entity Recognition	People, places, orgs, dates, quantities
Sentiment Analysis	Positive/negative/neutral + aspect-level opinions
Summarization	Concise summaries of long text
Key Phrase Extraction	Main concepts and key phrases
PII Redaction	Detects/redacts names, addresses, phone numbers
Language Detection	Identifies language
Text Analytics for Health	Medical entities (diagnoses, medications, symptoms)
Conversational Language Understanding	Intent/entity extraction from custom trained model
Custom Question Answering	Returns answers from a knowledge base

Language MCPTool in Code

```
from azure.ai.projects.models import MCPTool

mcp_tool = MCPTool(
    server_label="azure-language",
    server_url="https://{foundry-resource-name}.cognitiveservices.azure.com/language/mcp"
               "?api-version=2025-11-15-preview",
    require_approval="always",
)
# Optionally restrict: mcp_tool.allowed_tools = ["extract_named_entities_from_text"]
```

Client pattern:

```
openai_client = project_client.get_openai_client()
response = openai_client.responses.create(
    input=[{"role": "user", "content": user_prompt}],
    extra_body={"agent_reference": {"name": "Text-Analysis-Agent", "type": "agent_reference"}},
)
print(response.output_text)
```

Multi-task prompts: Agent can call multiple tools in one turn (e.g., NER + sentiment)

Speech with Generative AI

Speech-to-Text Models (OpenAI Audio API)

```
from openai import AzureOpenAI

client = AzureOpenAI(
    azure_endpoint=FOUNDRY_ENDPOINT, api_key=FOUNDRY_KEY,
    api_version="2025-03-01-preview"
)

audio_file = open("speech.mp3", "rb")
transcription = client.audio.transcriptions.create(
    model=MODEL_DEPLOYMENT,
    file=audio_file,
    response_format="text"
)
print(transcription)
```

Models: gpt-4o-transcribe, gpt-4o-mini-transcribe, gpt-4o-transcribe-diarize

Text-to-Speech Models (OpenAI Audio API)

```
with client.audio.speech.with_streaming_response.create(  
    model=MODEL_DEPLOYMENT,  
    voice="alloy",  
    input="This speech was AI-generated!",  
    instructions="Speak in an upbeat, excited tone.",  
) as response:  
    response.stream_to_file("speech_output.mp3")
```

Models: gpt-4o-tts, gpt-4o-mini-tts

Key parameters: voice (e.g., "alloy"), input (text), instructions (tone/style)

Azure Speech SDK

Choosing a Speech Capability

[EXAM]

Requirement	Choose
Transcribe a live phone call with low latency	Real-time speech-to-text on the streaming audio
Transcribe recordings that are already complete	Batch transcription
Two-way spoken interaction with low latency	Real-time speech-to-text in, text-to-speech out
Produce audio from text	Text-to-speech
Change the language of what was said	Speech translation

Batch transcription waits for the recording to finish, so it cannot serve a live transcript and it breaks turn-taking. Text-to-speech generates audio instead of transcribing it, translation changes language rather than producing a transcript, and embeddings support similarity search rather than conversation.

SpeechConfig and Speech-to-Text

[EXAM]

Package: `azure-cognitiveservices-speech` (SDK $\geq 1.48.2$)

```
import azure.cognitiveservices.speech as speech_sdk

speech_config = speech_sdk.SpeechConfig(subscription="KEY", endpoint="ENDPOINT")

audio_config = speech_sdk.audio.AudioConfig(filename="audio.wav")
# or: use_default_microphone=True

recognizer = speech_sdk.SpeechRecognizer(speech_config=speech_config,
                                          audio_config=audio_config)
result = recognizer.recognize_once_async().get()

if result.reason == speech_sdk.ResultReason.RecognizedSpeech:
    print(result.text)
```

ResultReason values: `RecognizedSpeech`, `NoMatch`, `Canceled`

SpeechRecognitionResult properties: `Duration`, `OffsetInTicks`, `Properties`, `Reason`, `ResultId`, `Text`

Text-to-Speech (SDK)

[EXAM]

```
audio_config = speech_sdk.audio.AudioOutputConfig(use_default_speaker=True)
speech_synthesizer = speech_sdk.SpeechSynthesizer(
    speech_config=speech_config, audio_config=audio_config
)
result = speech_synthesizer.speak_text_async("Hello world").get()

if result.reason == speech_sdk.ResultReason.SynthesizingAudioCompleted:
    print("Done")
```

Audio format:

```
speech_config.set_speech_synthesis_output_format(
    speech_sdk.SpeechSynthesisOutputFormat.Riff24Khz16BitMonoPcm
)
```

Voice:

```
speech_config.speech_synthesis_voice_name = 'en-US-Brian:DragonHDLatestNeural'
```

Custom Speech

[EXAM]

Train a custom speech-to-text model when base recognition mishears **product names, jargon, or accents**, then deploy it to a custom endpoint.

Workflow: create a project → upload audio and transcripts → train → evaluate → deploy to an endpoint

```
speech_config.endpoint_id = "<Custom Speech project ID>"
```

The REST request identifies the model by the **Custom Speech project ID** — not the project URL and not the endpoint URL.

When a custom model expires:

Behavior	Correct?
Requests fall back to the latest base model for that locale	Yes
Requests start returning 4xx errors	No
The expired model keeps serving until removed	No
The model is deleted automatically	No

Consequence: recognition keeps working, but domain accuracy silently drops. Monitor expiry dates and retrain before them.

SSML

[EXAM]

```
ssml = """
<speak xmlns:mstts="https://www.w3.org/2001/mstts" version="1.0" xml:lang="en-US">
  <voice name="en-US-JennyNeural">
    <mstts:express-as style="cheerful">
      Welcome to our service!
    <break time="500ms"/>
    How can I help you today?
  </mstts:express-as>
</voice>
</speak>
"""

result = speech_synthesizer.speak_ssml_async(ssml).get()
```

SSML capabilities: speaking style, pauses (<break>), phonemes (<phoneme>), prosody, say-as rules, recorded audio insertion

Requirement	Use
Pronounce a technical term or product name precisely	<phoneme>
Read dates, numbers, or currency correctly	<say-as>
Change rate, pitch, or volume	<prosody>
A brand-specific voice	Custom neural voice — requires training a model

Least effort for pronunciation is <phoneme>, not a custom neural voice.

Speech MCP Server

Speech MCP Endpoint and Tools

Endpoint:

<https://{foundry-resource-name}.cognitiveservices.azure.com/speech/mcp>

Two tools:

Capability	Details
Speech-to-text (Recognize)	Supports WAV, MP3, OGG, FLAC, MP4, M4A, AAC; language; phrase hints; profanity filtering (masked/removed/raw)
Text-to-speech (Synthesize)	Neural voices (e.g., en-US-JennyNeural); WAV/MP3 output; URL returned

Storage requirement: Azure Blob Storage account required for audio I/O

SAS URL permissions required: Read, Add, Create, Write, List

Speech MCPTool in Code

```
from azure.ai.projects.models import MCPTool

mcp_tool = MCPTool(
    server_label="azure-speech",
    server_url="https://{foundry-resource-name}.cognitiveservices.azure.com/speech/mcp",
    require_approval="always",
)
```

Portal connection settings:

- Foundry resource name
- Bearer (Ocp-Apim-Subscription-Key): project key
- X-Blob-Container-Url: SAS URL for blob container

Prompt-based customization: Voice selection (using the voice "en-GB-SoniaNeural"), language (es-ES), phrase hints, profanity filtering

Voice Live API

Why Real Time Is a Different Problem

Record → transcribe → think → speak takes several seconds. In a conversation, that feels broken.

Request/response (batch)

```
user speaks .....  
  wait  
  transcribe  
wait  
  model answers  
  wait  
play audio  
  
~3-6 seconds
```

Voice Live (streaming)

```
user speaks ....  
  ↓ audio streams up while they talk  
model starts answering  
  ↓  
audio streams back, chunk by chunk  
  
< 1 second, and interruptible
```

Interruption is the hard part. When `input_audio_buffer.speech_started` arrives, the user began talking over the answer — stop playback immediately.

VAD decides when a turn ended, so nobody has to press a button.

Voice Live — Real-Time Speech

[EXAM]

WebSocket-based, real-time bidirectional speech; low latency

Endpoints:

Connection type	Endpoint
Via Foundry project	<code>wss://{resource}.services.ai.azure.com/voice-live/realtime?api-version=2025-10-01</code>
Direct model	<code>wss://{resource}.cognitiveservices.azure.com/voice-live/realtime?api-version=2025-10-01</code>

Authentication (recommended): Entra ID Bearer token; requires **Cognitive Services User** role

API key auth: `api-key` header (not browser-compatible)

Why WebSocket: one connection carrying bidirectional streaming audio at roughly 100 ms latency.

Transport	Why not
RTMP	Built for one-way broadcast streaming
WebRTC	Peer-to-peer, not this client-to-service channel
SIP	Handles call signaling, not the audio stream itself

Key Events

Client → Server:

Event	Purpose
<code>session.update</code>	Modify session settings (voice, instructions, VAD)
<code>input_audio_buffer.append</code>	Add base64-encoded audio data
<code>response.create</code>	Trigger model inference

Server → Client:

Event	Purpose
<code>session.updated</code>	Session config confirmed
<code>response.done</code>	Response generation complete
<code>response.audio.delta</code>	Audio chunk for playback
<code>input_audio_buffer.speech_started</code>	User started speaking — cancel playback
<code>response.audio.transcript.done</code>	Transcript of AI response ready

Session Configuration

```
{
  "type": "session.update",
  "session": {
    "modalities": ["text", "audio"],
    "voice": { "type": "openai", "name": "alloy" },
    "instructions": "You are a helpful voice assistant.",
    "input_audio_format": "pcm16",
    "output_audio_format": "pcm16",
    "input_audio_sampling_rate": 24000,
    "turn_detection": {
      "type": "azure_semantic_vad",
      "threshold": 0.5,
      "prefix_padding_ms": 300,
      "silence_duration_ms": 500
    },
    "input_audio_noise_reduction": { "type": "azure_deep_noise_suppression" },
    "input_audio_echo_cancellation": { "type": "server_echo_cancellation" }
  }
}
```

Python SDK: `azure-ai-voicelive` (async-only, v1.0.0+)

Retrieval Practice — Lab 3

Pause here and complete [Lab 3](#), then check its solution.

Translation

Azure Translator — Text Translation

[EXAM]

90+ languages; package: `azure-ai-translation-text`

```
from azure.ai.translation.text import TextTranslationClient
from azure.ai.translation.text.models import InputTextItem
from azure.core.credentials import AzureKeyCredential

client = TextTranslationClient(credential=AzureKeyCredential("KEY"),
                              endpoint="FOUNDRY_ENDPOINT")
# or use region: TextTranslationClient(credential=..., region="FOUNDRY_REGION")

# Detect supported languages
languages = client.get_supported_languages(scope="translation")

# Translate (auto-detects source language)
result = client.translate(
    body=[InputTextItem(text="Hola"), InputTextItem(text="こんにちは")],
    to_language=["fr", "en"]
)
for doc in result:
    print(f"Detected: {doc.detected_language.language}")
    for t in doc.translations:
        print(f"  {t.to}: {t.text}")
```

Mixed-language input: split a segment containing multiple languages into single-language segments, then translate each segment separately. Automatic detection returns **one** language per segment, and forcing a single source language mistranslates the other one.

Transliteration and Speech Translation

Transliterate (script conversion):

```
result = client.transliterate(  
    body=[InputTextItem(text="こんにちは")],  
    language="ja", from_script="Jpan", to_script="Latn"  
)  
# → "Kon'nichiwa"
```

Speech Translation (SDK):

```
translation_cfg = speech_sdk.translation.SpeechTranslationConfig(  
    subscription="KEY", endpoint="ENDPOINT"  
)  
translation_cfg.speech_recognition_language = "en-US"  
translation_cfg.add_target_language("fr")  
translation_cfg.add_target_language("ja")  
  
translator = speech_sdk.translation.TranslationRecognizer(  
    translation_config=translation_cfg,  
    audio_config=speech_sdk.AudioConfig(use_default_microphone=True)  
)  
result = translator.recognize_once_async().get()  
print(result.translations["fr"])    # "Bonjour."  
print(result.translations["ja"])    # "こんにちは。"
```

Retrieval Practice — Lab 4

Pause here and complete [Lab 4](#) before continuing.

Azure AI Immersive Reader

[EXAM]

An assistive reading experience: read-aloud, line focus, syllable breaking, picture dictionary, and single-word translation.

Choose it when the requirement is **reading accessibility** — for example dyslexia support — with minimal setup.

Not this	Because it
Document Intelligence	Extracts content from documents
Azure Language	Analyzes text
Translator	Converts languages

None of them provide the assistive reading experience itself.

Private Networking for a Language Resource

[EXAM]

Analyzing confidential **on-premises** documents:

1. Deploy the **Language resource**
2. Add a **private endpoint** — the service gets a private IP inside your virtual network
3. Connect from on-premises over **VPN or ExpressRoute**

Alternative	Why it fails
Public endpoint	The service stays reachable over the internet
VPN or ExpressRoute alone	Without a private endpoint the service is not private
Managed identity	Controls authentication, not network routing

Identity answers “who”; networking answers “from where.”

Key Takeaways

1. **TextAnalyticsClient** — language detection, sentiment, NER, key phrases, entity linking; 5,120 char / 1,000 item limits
2. **Language MCP server** — 9 tools; agent decides which to use per prompt; `MCPTool` to attach
3. **OpenAI audio models** — `gpt-4o-transcribe/tts` for GPT-integrated speech pipelines; API version `2025-03-01-preview`
4. **Azure Speech SDK** — `SpeechConfig`, `SpeechRecognizer`, `SpeechSynthesizer`; SSML for style control
5. **Speech MCP server** — Blob Storage required for audio I/O; two tools: recognize + synthesize
6. **Voice Live API** — WebSocket real-time speech; `session.update` with VAD, noise reduction, echo cancellation
7. **Translation** — `TextTranslationClient`; `translate()` + `transliterate()`; `SpeechTranslationConfig` for spoken input
8. **Live vs recorded** — real-time speech-to-text for live audio, batch transcription only for completed recordings
9. **Custom NER** — fix low precision by narrowing the entity definition, not by lowering the threshold
10. **Immersive Reader** — the reading-accessibility service; **private endpoint + VPN/ExpressRoute** — the network-isolation pattern